

W. Shakespeare

Categories and functors:
much ado about nothing?

Bachelor thesis
26 April 1564

Thesis supervisors: prof.dr. G.A. Bredero
dr. J. van den Vondel



Leiden University
Mathematical Institute

Contents

1	Introduction	3
1.1	Assumptions	3
2	Relevant mathematics	4
2.1	Voronoi	4
2.2	Tabu search	4
2.3	Simulated annealing	4
3	General model	5
3.1	SAMP	5
3.2	Network structure	5
3.3	Transit Assignment	6
3.4	Costs	7
3.5	Access	7
3.6	The model	9
3.7	General algorithm	9
4	Applied model	10
4.1	Chosen parameters	10
4.2	Difference in choices	10
5	Acquiring data	11
5.1	Busses	11
5.1.1	Buslines	11
5.1.2	Busschedule	11
5.1.3	Busstops	11
5.2	Destination points	11
5.3	Population	12
5.3.1	PC4	12
5.3.2	Voronoi regions	12
6	Explanation of experiment	13
7	Results	13
8	Discussion	13

1 Introduction

1.1 Assumptions

The following things are assumed to be true

1. The group of people that the model is made for is very small compared to daily travelers, therefore this small group will have no effect on overcrowdedness or other usage related metrics.

2 Relevant mathematics

2.1 Voronoi

2.2 Tabu search

2.3 Simulated annealing

3 General model

This section summarizes the general model and algorithm developed by Rumpf and Kaul in [RK21]. As the model and algorithm is made by them, a more detailed explanation can be found in their paper. In this section we will give a summary so the later parts of this study can be understood.

3.1 SAMP

The algorithm made by Rumpf and Kaul in [RK21] was made to solve the *social access maximization problem* (SAMP). The formulation of the problem is very general as to be applicable to many situations.

Let L be the set of considered transitlines, where a transitline $l \in L$ is a series of stops with a given traveltime between each stop. A schedule y is a vector with $y_l \in \mathbb{N}$ the amount of vehicles assigned to line l in a given time period. The vehicles are distributed evenly over the time period, so 5 vehicles in an hour gives 1 trip every 12 minutes.

The initial schedule is denoted as y^* . The model has upper and lower bounds $y_l^{\min} \leq y_l \leq y_l^{\max}$ for all $l \in L$. These can be set to 0 and infinity if no restriction is needed, or both to y_l^* if no change is wanted.

Let x be the user flow vector for the transit network, x^* the initial flow vector. User flow represents how busy parts of the network are, which is influenced by the schedule y and found by the $\text{TransitAssignment}(y)$ function, which will be explained further in Section 3.3.

Other restraints of the model are given by operating- and user costs with upper bounds $B_{operator}, B_{user}$ respectively. Operating costs are mostly made up by monetary costs of running the transit network and monetary gain of users paying to use the service. User costs are made up of difficulty in traversing the network, be that in monetary cost or time spent in the system. More on these costs in Sections 3.4.

Let $\text{Access}(y)$ be a function describing the social access objective. This value only depends on the design of the network y and not on user flow x , because the model is designed for a small group of individuals following assumption 1. Different choices for $\text{Access}(y)$ are given in 3.5.

3.2 Network structure

The model uses a representation of the transit network to calculate traveltime and routechoices. For this, a directed graph $G = (V, A)$ is used. V is the set of all relevant nodes, be that origin-, destination-, transitstopnodes. A is the set of arrows linking the nodes together, these represent different modes of travel like walking, bus- or trainriding.

The node set is partitioned in subsets, each containing specific nodes. Let $V_{stop} \subset V$ contain nodes representing the physical location of transitstops. Let $V_{board} \subset V$ be the set of boarding nodes. Each stopnode $i \in V_{stop}$ has boardingnodes $j \in V_{board}$ for every line $l \in L$ that stops at that stop. Boarding

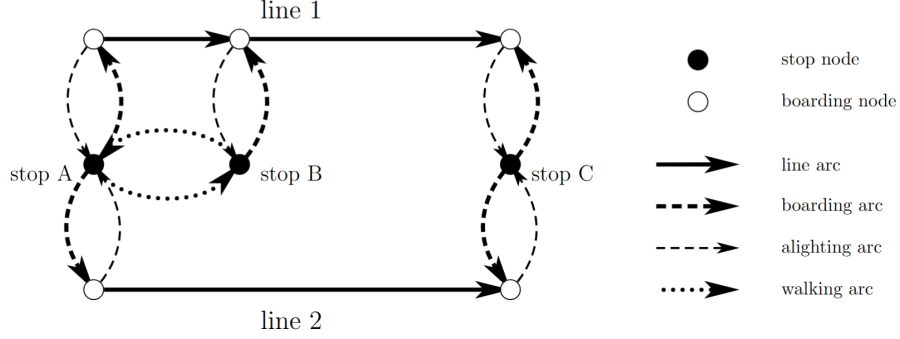


Figure 1: Example of network representation containing two lines and three stops. Each stop has nodes in V_{stop} . Line 1 stops at all stops and thus has three boarding nodes, line 2 only has boarding nodes at stop A and stop C. Boarding nodes are connected to their stops with boarding arcs and alighting arcs. Boarding nodes are connected by line arcs if a transitline connects their respective stops. Stop A and stop B are connected by walking arc, because they are close enough. Figure taken from [RK21].

nodes are used to distinguish between transitlines. Let $V^+, V^- \subset V$ be origin and destination nodes, respectively. These represent the physical places where trips begin and end. To calculate social access, subsets $\tilde{V}^+, \tilde{V}^- \subset V$ are added, representing community and facility nodes, respectively.

Just like the node set, the arc set A is partitioned in subsets. Let $A_{line} \subset A$ be the arcs that represent the journey made by transitlines and let A_{board} , $A_{alight} \subset A$ be the arcs representing entering or leaving a transit vehicle at a certain stop. Let $A_{walking} \subset A$ be walking arcs, connecting different nodes if their physical distance is walkable.

3.3 Transit Assignment

The transit assignment model, $\text{TransitAssignment}(y)$, models the way daily users interact with a given schedule y . It assumes users try to take their optimal route, which may not be the fastest due to overcrowdedness. The flow produced by the model is called *user-optimal flow* and is often less optimal than when all flow is controlled directly.

There are many such models, but Rumpf and Kaul choose the nonlinear model by Spiess and Florian in [SF89]. This model increases arc cost based on crowdedness, discouraging flows from exceeding line capacity. The user-optimal flow output can not be changed in a way that individual users benefit from changes in only their route. The model is relatively simple, a desired property, as the model is run many times during the algorithm.

3.4 Costs

Generally, operator costs are monetary costs relevant to running the transit system. Things like purchasing and maintaining vehicles, employing drivers and other staff and other costs. In this study, we restrict the total number of busses to not increase. This gives no reason to purchase new vehicles or employ more drivers, so we assume that operator costs will not increase with a new schedule. This means that we can omit the operator cost constraint from the model.

User costs are time costs made by using the system. Things like travel time, wait time, number of transfers and discomfort. All the user costs can be weighted based on transfer mode, 5 minutes of walking may weigh more than 5 minutes of busriding for example. For each $(i, j) \in A$ a flow amount x_{ij} and cost c_{ij} can be implemented. Flow amount represents the amount of users taking that arc, while cost represents some penalty in using that arc, like time needed to traverse it. The user cost function takes the form

$$\text{UserCost}(x, \omega) = \theta_1 \sum_{ij \in A_{line}} c_{ij} x_{ij} + \theta_2 \sum_{ij \in A_{walk}} c_{ij} x_{ij} + \theta_3 \omega,$$

where $\omega \in \mathbb{R}$ is the total time spent waiting at busstops by all users combined.

Due to assumption 1 we do not want the total user cost to increase too much, so we introduce a parameter ε to give a constant upper bound

$$\text{UserCost}(x, \omega) \leq (1 + \varepsilon) \text{UserCost}(x^*, \omega^*).$$

When choosing ε , a balance needs to be found between a large increase, giving a larger set of possible solutions and possibly a more optimal one at the cost of daily user experience.

3.5 Access

The objective function $\text{Access}(y)$ should describe how easy it is to travel from a general community to a general facility, with a focus on equity of access. To do this, a metric is made to measure the accessibility for a specific community $i \in \tilde{V}^+$.

The chosen metric $A_i(y)$ is *gravity-based*, meaning that the attractiveness of facilities reduces gradually as distance increases. An alternative is a discrete distance based approach, where facilities $j \in \tilde{V}^-$ are considered only if they are within a certain distance from the source community i . This approach is not preferred, as facilities that lie just inside the considered radius are weighted equally to facilities that are nextdoor, and facilities that lie just a bit further are not considered at all. This is solved by including a distance function $d_{ij}(y)$, measuring time traveled from i to j , and a gravity decay parameter $\beta > 0$. A larger β will result in a higher penalty for further travel.

As the users can choose between facilities, a competition-related metric $F_j(y)$

is used to find the crowdedness of facilities. It is defined as

$$F_j(y) = \sum_{k \in \tilde{V}^+} P_k d_{kj}^{-\beta}(y) \quad \forall j \in \tilde{V}^-$$

where P_k is the population of community k and $d_{kj}^{-\beta}(y)$ is the previously mentioned gravity decay. This crowdedness can be compared to the quality S_j of a facility, usually taken as some metric of patient capacity.

The accessibility metric for a given region is then defined as

$$A_i(y) = \sum_{j \in \tilde{V}^-} \frac{S_j d_{ij}^{-\beta}(y)}{F_j(y)} \quad \forall i \in \tilde{V}^+.$$

This metric was first developed in [Wei76] to describe differential access to employment opportunities, but can also be used for other competition based access cases.

Since the goal is to create more equity in accessibility, not all communities should be included in the overall objective function. Instead, the communities with the worst accessibility should be improved. By maximizing the minimum communities, a more equitable scenario is achieved. The model parameter $\mathcal{K}$ is introduced to achieve this. The objective function is then defined as

$$\text{Access}(y) = \sum_{\substack{\mathcal{K} \text{ minimum elements of} \\ \{A_i(y) | i \in \tilde{V}^+\}}} A_i(y).$$

Note that the objective function will be evaluated many times, and that the $\mathcal{K}$ worst communities are dependent on the y of the current iteration, so the evaluated and improved communities changes while the algorithm is running.

The parameter $\mathcal{K}$ needs to be chosen between 1 and $|\tilde{V}^+|$, where extreme values are not beneficial to the outcome. Low values for $\mathcal{K}$, like 1, may result in minor improvements to a single community with no regard for the relatively large effect on other communities. High values, like $|\tilde{V}^+|$, may result in making large improvements to the highly accessible communities at a cost of making the less accessible communities even worse, as long as the total accessibility increases.

3.6 The model

The SAMP model takes the shape of the following nonlinear program

$$\max_y \text{Access}(y) \quad (1)$$

$$\text{s.t.} \quad \sum_{l \in L} y_l \leq \sum_{l \in L} y_l^* \quad (2)$$

$$y_l^{\min} \leq y_l \leq y_l^{\max} \quad \forall l \in L \quad (3)$$

$$y_l \in \mathbb{N} \quad \forall l \in L \quad (4)$$

$$x = \text{TransitAssignment}(y) \quad (5)$$

$$\text{UserCost}(x, \omega) \leq (1 + \varepsilon) \text{UserCost}(x^*, \omega^*) \quad (6)$$

Where (1) is the objective function described in section 3.5, (2) constrains the total number of vehicles to not increase from the initial amount, (3) ensures upper and lower bounds for vehicle allocation, (4) ensures a whole number of vehicles is allocated to each line, (5) defines user flow as discussed in section 3.3 and (6) gives an upper bound to the increase in user cost.

3.7 General algorithm

4 Applied model

4.1 Chosen parameters

4.2 Difference in choises

5 Acquiring data

To run the algorithm, some data needs to be acquired. This section will provide the methodology of how this data was acquired, full tables of data can be found in the appendix.

5.1 Busses

5.1.1 Buslines

Only buslines that travel at least partially through the study area are taken into account for this study. Lines that are not entirely inside the study area are truncated to only have stops within it. The stop were such lines enter or leave the municipality are treated as starting- or ending stops, respectively. This gives that some lines that differ outside the municipality appear identical in our study.

5.1.2 Busschedule

After the buslines were selected, a schedule was retrieved from [Qbu25]. The chosen schedule was for monday the 10th of february of 2025, between 14:00 and 15:00 in the afternoon. This weekday and timeframe was chosen as a ordinary, non-rush hour time, that represents ordinary bus travel.

The information given by a schedule is average traveltime between stops, wait time at a stop and amount of busses per line per hour. The algorithm used takes busses per line per timeframe and averages this out to arrive at the frequency of a line, where more busses per timeframe would mean a higher frequency of service.

5.1.3 Busstops

Traveltime between busstops was extracted from the schedule, as was their name. The other information needed was their location, so that distances between origin or destination points can be found and walking arcs could be connected between busstops, GP offices and community centers. As most busstops consist of a pair of physical locations, one for each direction, their midpoint is taken as the location of the pair overall.

5.2 Destination points

The destination points are given by the location of general practitioners (GP's) in Leiden. The GP-offices have a location and a quality metric. A full list of GP's in Leiden can be found on [Ned25], including a link for each individual office's website. This also gives the location and amount of full-time employed doctors. The latter was taken as the quality of a GP-office, where more doctors would imply a higher capacity of patients and thus a higher quality.

5.3 Population

All trips originate from a community center. Each community center is a point, representing a region. All trips within a region are assumed to originate from that region's community center. From this center, walking arcs are made connecting nearby bus stops. Each community center has a location and a population. The location determines which busstops can be connected by a walking arc and the population is used by the algorithm to determine the amount of trips originating from a community. Two methods of dividing the municipality into regions are used and compared, namely the PC4 regions and voronoi regions.

5.3.1 PC4

Each address in the Netherlands has a postcode, consisting of four numbers and two letters. If one only uses the four numbers, the Netherlands can be divided into pc4 regions. This division roughly resembles which areas were built at the same time.

The location of each community center is taken as the rough midpoint of the region, and the population is taken from [All25].

5.3.2 Voronoi regions

According to [Wan22], travelers prefer to take a bus at the nearest bus stop, even when that involves a transfer instead of a direct route. To apply this effect on our study, the regions we choose should reflect trips that start at the closest bus stop. A voronoi diagram is the ideal method for this.

A voronoi diagram is created by using a set of weightpoints in a plane, we use the busstops as weightpoints. Via an algorithm, the plane is divided into polygons. Each polygon contains exactly one weightpoint and all points in a polygon have that weightpoint as their closest weightpoint.

When the plane is infinite, the outer voronoi regions will also be infinite. For this study, a border had to be drawn to limit all voronoi regions to be within the municipality of Leiden.

We again need a community center location and a population. Since each bus stop is taken as a weightpoint, these also serve as the community center location. To approximate the population of each voronoi region we estimate the overlap between a voronoi- and a pc4 region. Let

- V set of voronoi regions,
- P set of postcode regions,
- s_v surface area for $v \in V$,
- d_p population density for $p \in P$,
- $f(v, p)$ fraction of v that lies in p .

Then, the population of a voronoi region v is given by

$$s_v \sum_{p \in P} d_p f(v, p).$$

6 Explanation of experiment

7 Results

8 Discussion

- intro
- math theory (voronoi, tabu search)
- Rumpf kaul general model (with application of chicago)
- application on Leiden (with differences)
- data used
- results

References

- [All25] Allecijfers. May 1, 2025. URL: <https://allecijfers.nl/postcode/>.
- [Ned25] Patientenfederatie Nederland. May 2, 2025. URL: <https://www.zorgkaartnederland.nl/huisartsenpraktijk/leiden>.
- [Qbu25] Qbuzz. Feb. 3, 2025. URL: <https://zhn.qbuzz.nl/?direction=1&date=2025-02-10>.
- [RK21] Adam Rumpf and Hemanshu Kaul. “A public transit network optimization model for equitable access to social services”. eng. In: (2021).
- [SF89] Heinz Spiess and Michael Florian. “Optimal strategies: A new assignment model for transit networks”. In: *Transportation Research Part B: Methodological* 23.2 (1989), pp. 83–102. ISSN: 0191-2615. DOI: [https://doi.org/10.1016/0191-2615\(89\)90034-9](https://doi.org/10.1016/0191-2615(89)90034-9). URL: <https://www.sciencedirect.com/science/article/pii/0191261589900349>.
- [Wan22] M. Wang F.; Ye. “Optimization Method for Conventional Bus Stop Placement and the Bus Line Network Based on the Voronoi Diagram.” In: *Sustainability* 14.7918 (2022). DOI: <https://doi.org/10.3390/su14137918>.
- [Wei76] Jörgen W. Weibull. “An axiomatic approach to the measurement of accessibility”. In: *Regional Science and Urban Economics* 6.4 (1976), pp. 357–379. ISSN: 0166-0462. DOI: [https://doi.org/10.1016/0166-0462\(76\)90031-4](https://doi.org/10.1016/0166-0462(76)90031-4). URL: <https://www.sciencedirect.com/science/article/pii/0166046276900314>.